

MID IISolution-Artificial Intelligence (AI-2002)

Question 01 (1 mark each)

- Ordering decides the sequence of decisions (0.25). Filtering removes invalid options (0.25). Forward Checking “Look ahead” filtering to avoid future conflicts. (0.5)
- Strengths:** Fast, simple, works well for text and high-dimensional data (0.4)
Weakness: Unrealistic independence assumption + zero probability issue (0.4)
 Solution: *Feature Selection / Bayesian Networks* → mitigates independence assumption (0.2)
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Aspect	Mini-Max	Alpha-Beta Pruning	
Worst Time	$O(b^d)$	$O(b^d)$	(0.2 marks)
Best Time	$O(b^d)$	$O(b^{\lceil d/2 \rceil})$	(0.4 marks)
Space	$O(b \cdot d)$	$O(b \cdot d)$	(0.4 marks)

Question 02: Part (a)

- Yes = 6, No = 6 → $P(\text{Yes}) = P(\text{No}) = 6/12 = 0.5$; Yes, class (rows 1, 3, 4, 7, 9, 11 — six total): (1 mark)
 - Age=Old: rows 1, 3, 7, 9, 11 → $5/6 = 0.833$
 - Cholesterol=High: rows 1, 3, 4, 7, 11 → $4/6 = 0.667$ (row 9 is Low)
- No class (rows 2, 5, 6, 8, 10, 12 — six total): (1 mark)
 - Age=Old: rows 5, 12 → $2/6 = 0.333$
 - Cholesterol=High: row 8 only → $1/6 = 0.167$
- Posteriors for new patient (Old, High): (1 mark)
 - $P(\text{Yes} \mid \text{Old, High}) \propto 0.5 \times 0.833 \times 0.667 = 0.278$
 - $P(\text{No} \mid \text{Old, High}) \propto 0.5 \times 0.333 \times 0.167 = 0.028$

Since $0.278 > 0.028 \rightarrow \text{Heart Disease} = \text{Yes}$

Question 02: Part (b)

- a) $P(C) \cdot P(L|C) \cdot P(\neg M|C) \cdot P(\neg B|L, M)$ (1.5 marks)
- $P(C = \text{true}) = 0.1$
 - $P(L = \text{true} \mid C = \text{true}) = 0.2$
 - $P(M = \text{true} \mid C = \text{true}) = 0.4 \Rightarrow P(\neg M \mid C) = 0.6$
 - $P(B = \text{true} \mid L = \text{true}, M = \text{false}) = 0.9 \Rightarrow P(\neg B) = 0.1$

Answer: 0.0012

- b) Enumeration: Needs all combinations → slow (0.5 marks)
 Bayesian Network: Uses independence and Fewer computations
 Conclusion: Bayesian networks reduce exponential complexity to efficient factorized computation

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Question 02: Part (c) (0.5 each)

$$P(\text{Graduate} \mid \text{Online}) = \frac{P(\text{Graduate} \cap \text{Online})}{P(\text{Online})}$$

$$= \frac{0.12}{0.35} = 0.3429$$

1)

$$P(\text{Hybrid} \mid \text{Undergraduate}) = \frac{0.10}{0.50}$$

$$= 0.20$$

2)

First compute:

$$P(\text{PhD} \mid \text{OnCampus}) = \frac{0.03}{0.45} = 0.0667$$

3)

So,

$$P(\text{Not PhD} \mid \text{OnCampus}) = 1 - 0.0667 = 0.9333$$

4)

First find:

$$P(G \cup \text{PhD}) = 0.40 + 0.10 = 0.50$$

Online among these groups:

$$P(O \cap (G \cup \text{PhD})) = 0.12 + 0.05 = 0.17$$

Now:

$$P(O \mid G \cup \text{PhD}) = \frac{0.17}{0.50} = 0.34$$

Question 3: Part (a)

Variables: 0.25
Domains: 0.25
Constraints: 0.5
Scheduling: 1.25

Constraints:

- 3 Days $\times$ 3 Shifts = **9 total assignments**
- Nurses: **N1, N2, N3, N4**
- Each shift $\rightarrow$ **exactly 1 nurse**
- Each nurse $\rightarrow \leq 1$ **shift per day**
- **N1** $\times$ **Night shifts**
- **N2** ≥ 2 **Morning shifts**
- **N3 & N4** $\times$ **same shift on same day**

Day	Morning (M)	Evening (E)	Night (N)
Day 1	N2	N3	N4
Day 2	N2	N1	N3
Day 3	N3	N4	N2

b) Yes, multiple solutions exist. (0.25)

Justification:

- This is a **Constraint Satisfaction Problem (CSP)** with:
 - Multiple variables (9 shifts)
 - Flexible domains (4 nurses)
- Constraints **restrict but do not fully determine** assignments
- Different valid permutations exist:
 - Swap N3/N4 across days (while respecting constraint)
 - Reassign N2's required morning shifts across different days
 - Use N1 in different non-night slots

Hence, the solution space is **not unique**, and **many valid schedules** can satisfy all constraints.

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Question 03: Part (b)

Pruned Branches shown properly: 1
 Values shown at node properly: 1
 Traversing shown properly: 0.5

